Lecture 2 Homework

Tuesday, March 10, 2026 9:38 PM

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Homework Practice Sheet (Intro ML / CNNs)
Convolutions, Output Shapes + Concepts

Instructions

Show your work. Unless otherwise stated, use the CNN convention for convolution layers (*no kernel flipping*; i.e. cross-correlation as used in practice).

For a convolution with input height/width $H \times W$, kernel size K, padding P, and stride S, use:

$$H_{\text{out}} = \left\lfloor \frac{H + 2P - K}{S} \right\rfloor + 1, \qquad W_{\text{out}} = \left\lfloor \frac{W + 2P - K}{S} \right\rfloor + 1.$$

For conceptual questions, answer in 1–3 sentences.

Part A: Numeric Problems (CNN Basics)

1. **Convolution output dimensions.**
An input image has shape $8 \times 8 \times 1$. A convolution layer uses:

$$\text{kernel size } 3 \times 3, \qquad \text{stride } 1, \qquad \text{padding } 0,$$

with **4 filters**. Compute the output shape.

$H_{out} = \left\lfloor \frac{8+0-3}{1} \right\rfloor + 1 = 6$

$W_{out} = \left\lfloor \frac{8+0-3}{1} \right\rfloor + 1 = 6$

Output shape: $6 \times 6 \times 4$

2. **Convolution output dimensions with padding.**
An input image has shape $10 \times 10 \times 1$. A convolution layer uses:

$$\text{kernel size } 3 \times 3, \qquad \text{stride } 1, \qquad \text{padding } 1,$$

with **2 filters**. Compute the output shape.

$H_{out} = \left\lfloor \frac{10+2-3}{1} \right\rfloor + 1 = 10$

$W_{out} = \left\lfloor \frac{10+2-3}{1} \right\rfloor + 1 = 10$

Output shape: $10 \times 10 \times 2$

3. **Number of parameters in a convolution layer.**
A convolution layer takes an input with **1 channel** and uses **3 filters**, each of size 3×3. Assume each filter has its own bias term.
How many trainable parameters are in this layer?

$P = (3 \times 3 \times 1 + 1) \times 3 = 30$ trainable parameters

filter, input channel, filter, output channel

4. **Small convolution by hand.**
Let the input feature map be

$$X = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 2 & 1 & 0 \end{bmatrix}$$

and let the kernel be

$$K = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Using stride 1 and padding 0, compute the full output feature map.

Full output feature map $= \begin{bmatrix} 2 & 3 \\ 1 & 1 \end{bmatrix}$

5. **Max-pooling.**
Apply 2×2 max-pooling with stride 2 to

$$\begin{bmatrix} 1 & 3 & 2 & 0 \\ 4 & 2 & 5 & 1 \\ 0 & 2 & 7 & 3 \\ 1 & 6 & 4 & 2 \end{bmatrix}.$$

Compute the output.

Output $= \begin{bmatrix} 4 & 5 \\ 6 & 7 \end{bmatrix}$

Part B: Conceptual Questions (Short Answers)

6. **Why do CNNs use filters?**
In simple terms, what is a filter in a CNN trying to detect?

A filter is trying to detect specific patterns / features like edges or drastic changes in the pixels.

7. **What does padding do?**
What is the purpose of padding in a convolutional layer?

The purpose of padding is to ensure the edges of the input is not lost & included.

8. **What does max-pooling do?**
Why might we use max-pooling in a CNN?

We use it to reduce the size of the representation & to allow for translation invariance, since even though you lose detail, it keeps the key points so if something shifts, it will still be recognized as the same thing before the shift.